# Reciprocity in Lying for Others*

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**Abstract**

In an experimental setup simulating peer-review (i.e. the evaluation of others' performance in work environments), we investigate the impact of reciprocity. Our formal model predicts that in sequential assessment settings, reciprocal behaviour prompts agents to systematically over-report. A series of online experiments substantiate this prediction, revealing that the conduct of first movers diverges markedly from truthful reporting, leaning distinctly towards maximal over-reporting. Notably, this tendency weakens in conditions simulating anonymous/simultaneous interactions, indicating that the structure of peer-reviewing is key to ensuring the integrity of the system.

**Keywords: Reciprocity, Lying, Gift Exchange**

**JEL Classification: C7, C9, D9**

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# 1 Introduction

The prevalence of deception in various work environments, highlighted by recent scandals in finance, the automobile industry, sports, and academia, underscores the critical need to comprehend its dynamics. An established fact is that people's approach to deception is intricately linked to the extent to which it impacts others, even in environments without associated penalties for lying (Gneezy, 2005; Gerlach et al., 2019; Erat and Gneezy, 2012). Specifically, individuals demonstrate a heightened tendency to deceive when it proves beneficial for others, motivated by factors such as pure altruism, inequity aversion (Fehr and Schmidt, 1999), or reciprocity (Levine, 1998).

This study focuses on a unique form of reciprocity, where individuals may engage in deceptive behaviour on behalf of others (i.e. agents' reports exclusively influence the payoffs of others, not their own) *with the anticipation of future reciprocation*. Indeed, peer-review often necessitates workers to take turns evaluating each other's performance. This may introduce incentives for lying, distorting the assessments of others' performance. The implications for fields like academic research, where peer review is crucial, are obvious. While nominally anonymous, practical anonymity is challenging to maintain, especially in smaller scientific domains. If the empirical validity of lying and expecting reciprocal benefit holds, systems like peer review could face important distortions.

The simple environment under consideration involves two individuals, each tasked with assessing someone else. By varying the sequence between assessments (simultaneous vs. sequential) and the nature of the relationship between players (belonging to a group or not, and assessing the assessor or not), we can examine the performance of the model and explore the role of reciprocity in our setting. Our theory predicts that in an environment with sequential and reciprocal assessments among the two agents, the first-moving agent will be inclined to provide a higher assessment of the second player, anticipating a higher assessment in return. We examine our predictions in a set of online experiments with participants representative of the UK and the US population. Our experiments were pre-registered at the American Economic Association RCT registry.

We inform our experimental design by analysing first a formal model of reciprocity in assessments, within a framework of interacting altruism levels (Levine, 1998). In such an environment, lying-averse assessors, characterized by a privately observed altruism parameter, will judge the quality/performance of an another agent differently depending: 1) on their own altruism, and 2) on the perceived altruism of the agent whose performance they assess. That is, agents want to be more generous to more altruistic individuals, and hence if an assessor expects to be assessed in the future by the agent she assesses now, she has incentives to over-report and thus exaggerate her altruism level. When reciprocity in assessment is not present, then this channel is closed and the report only depends on the performance and the level of altruism of the assessor.

In the experiments, the actual "quality" of individual performance is determined using a roll of a six-sided dice observed by the assessor, but individuals are free to report any value from zero to five. Disentangling the performance under assessment from decisions that may be affected by the applied peer-review procedure, insulates our analysis from relevant endogeneity concerns. Indeed, if subjects were asked to perform a task that would be subsequently assessed, then the details of the assessment process could influence their performance, confounding the performance indicator in multiple non-trivial ways. Our design controls for the quality/performance under assessment, while also ensuring its non-verifiability to everyone except the assessor which is key for our purposes.

Our results reveal that, indeed, participants are willing to lie (i.e. report an excessively good assessment) to induce a high assessment in return. We thus receive some support for the model's predictions. Additionally, first movers disproportionately report the maximum allowable value for others. This can be construed as an attempt to signal clearly that they are willing to lie to benefit the other person. Moreover, average reports are consistently higher than the (unbiased) expected report. Interestingly, in the sequential and reciprocal environment, we find a positive correlation between the reports of second movers and the reports of first movers (see Table 5). That is, second movers appear to respond to first movers' reports, providing, to the best of our knowledge, the first pieces of evidence of reciprocity in lying within peer-review systems.

Most other experiments in the literature (Gneezy, 2005; Erat and Gneezy, 2012; Wiltermuth, 2011) do not consider reciprocal opportunities to lie for the other person. Instead they examine various forms of splitting the surplus between the agent that can potentially lie, and others. Colzani et al. (2023) also employ the Fischbacher and Föllmi-Heusi (2013) paradigm and have participants matched in a simultaneous vs. a sequential treatment. However, participants' report affect their own payoffs, rather than the payoffs of the other person. Ours is the first theoretically-driven empirical study in the potential of positive reciprocity in lying behaviour. Alempaki et al. (2019) also consider the issue of whether lying can be used as a device for reciprocity, but the domain of reciprocity is different from ours: they have an initial task that involves splitting the pie in a dictator experiment, while we have a task that involves potential lying at both stages of the game. In addition, our predictions are supported by a formal model of reciprocity. Dato et al. (2019) also have a model of reciprocity that adapts Levine (1998) and they consider the effect of reciprocity on lying behaviour. However, their setting is different. They consider interactions with negatively correlated payoffs, whereas our interest is in an environment where people could lie to benefit others, rather than themselves. Instead of a competitive setting with negative reciprocity, we consider an environment with potentially positive reciprocity. Importantly, our focus is on the optimal behaviour of the first mover, and how it differs across environments where the reciprocal relationship can be present or absent (capturing different assumptions about the size of the underlying population, and therefore the probability of repeated play).

## 2 The Model

Our model is based on the influential framework of Levine (1998), adapted to the environment of reciprocal assessments. There are two players $P_1$ and $P_2$, each observing a private draw from a known distribution. Unlike the standard die-rolling paradigm in which a player's payoff depends on their own report of the private draw, each player's payoffs depend on the other player's report in our model. As we shall see, for the players who move first, the prospect of inducing reciprocity by indicating that one is of an altruistic type will matter for the assessment of the other person.

***Private Draws and Reports*** $P_1$ observes a private draw $s_2$ from distribution $\mathcal{N}(\mu_{s_2}, \sigma^2_{s_2})$ and she reports $x$ (that is, they claim that $s_2 = x$). This report determines the payoff of $P_2$, which is equal to that report. Similarly, $P_2$ observes a private draw $s_1$ from $\mathcal{N}(\mu_{s_1}, \sigma^2_{s_1})$, and she reports $y$, which in turn becomes the payoff of $P_1$. Therefore, both players can potentially affect their opponents' payoffs by lying, i.e., by reporting an $x$ ($y$) that is not equal to $s_2$ ($s_1$) but higher or lower.

***Reciprocal Utility*** In addition to the monetary payoffs ($x$ for $P_2$, and $y$ for $P_1$), players receive an additional component, namely "reciprocal utility". That is, they derive utility not only from their own payoff but also from the payoff of the other player. The key determinants of an agent's reciprocal utility are the two players' reciprocity coefficients. For player $P_i$, the reciprocity coefficient is defined to be $\alpha_i \sim \mathcal{N}(\mu_{\alpha_i}, \sigma^2_{\alpha_i})$. A positive value of this coefficient simulates altruism and a negative value simulates spite. The larger the absolute value of this parameter the higher the level of altruism/spite.

Importantly, in the setting of Levine (1998) an agent's reciprocal utility is determined both by own reciprocity coefficient and by the (expected) reciprocity coefficient of the other player. That is, an agent derives more reciprocal utility when she is more altruistic herself, but also when she interacts with a more altruistic agent.[1]

The key predictions of our model concern behaviour in the 'simultaneous moves' vs. 'sequential moves' case.

## Simultaneous setting

In the case of $P_1$ and $P_2$ moving simultaneously, they do not observe their opponents' reports, and thus $x$ and $y$ are independent. $P_1$ solves the following maximisation problem,

$$\max_x \quad U(x) = -(s_2 - x)^2 + (\alpha_1 + E(\alpha_2))x + E(y)$$

where $-(s_2 - x)^2$ represents $P_1$'s cost of lying and $(\alpha_1 + E(\alpha_2))x$ is the "reciprocal utility" that $P_1$ receives from the interaction. Intuitively, the higher the expected sum of reciprocity components, the happier $P_1$ becomes by choosing a high report $x$ for the other person. Importantly, $P_1$ has no posterior information on the basis of which to estimate $P_2$'s reciprocity, and thus he bases his decision on $E(\alpha_2) = \mu_{\alpha_2}$. Turning to $P_2$'s maximisation problem, we observe that it is similar,

$$\max_y \quad U(y) = -(s_1 - y)^2 + (\alpha_2 + E(\alpha_1))y + E(x)$$

where $-(s_1 - y)^2$ is $P_2$'s lying cost and $(\alpha_2 + E(\alpha_1))y$ is her "reciprocal utility". It is straightforward to calculate the first-order conditions and the equilibrium reports are as follows,

$$\begin{cases} x^* = \frac{1}{2}(\alpha_1 + \mu_{\alpha_2} + 2s_2) \\ y^* = \frac{1}{2}(\alpha_2 + \mu_{\alpha_1} + 2s_1) \end{cases} \tag{1}$$

These conditions indicate that the mere presence of reciprocal utility makes

[1] Levine (1998) considers binary types, i.e. altruists and non-altruists, while we allow for a richer type space that nests varying levels of altruism and spite.

reports consistently greater that the average draw (in expectation).

## Sequential setting

Players move sequentially with $P_1$ being the first mover and $P_2$ being the second mover. In this case, $x$ and $y$ are dependent, as $P_2$ observes $x$ before she reports for $P_1$ and $P_1$ knows about it. Therefore, $P_1$ solves the following maximisation problem,

$$\max_x \quad U(x) = -(s_2 - x)^2 + (\alpha_1 + E(\alpha_2))x + E(y|x)$$

where $-(s_2 - x)^2$ represents the cost of lying for $P_1$, $(\alpha_1 + E(\alpha_2))x$ is her reciprocal utility, and $E(y|x)$ is $P_2$'s expected report $y$ conditional on $x$, i.e., $P_1$'s expected payoff conditional on $P_2$'s report. That is, $P_1$ now can condition the expected report of her opponent on her own report (and given the opponent's assumed strategy). As for $P_2$, she observes $x$, and thus she knows her exact payoff, and her expectation of $P_1$'s reciprocal coefficient is conditioned on $P_1$'s report. $P_2$'s maximisation problem is as follows,

$$\max_y \quad U(y) = -(s_1 - y)^2 + (E(\alpha_1|x) + \alpha_2)y + x.$$

Assume a linear equilibrium exists, where $P_1$'s expected payoff in equilibrium is a linear function of $x$, given by $bx + z$. Then, $P_1$'s maximisation problem becomes,

$$\max_x \quad U(x) = -(s_2 - x)^2 + (\alpha_1 + E(\alpha_2))x + bx + z$$

and it's first-order condition yields,

$$\begin{aligned} x &= \frac{1}{2}(\alpha_1 + b + E(\alpha_2) + 2s_2) \\ \iff 2x - b - \mu_{\alpha_1} - \mu_{\alpha_2} - 2\mu_{s_2} &= \alpha_1 - \mu_{\alpha_1} + 2s_2 - 2\mu_{s_2} \end{aligned}$$

which can be used to rewrite $P_2$'s expectation of $P_1$'s reciprocal coefficient,

$$\begin{aligned}
E(\alpha_1|x) &= E(\alpha_1 - \mu_{\alpha_1}|x) + \mu_{\alpha_1} \\
&= E(\alpha_1 - \mu_{\alpha_1}|2x - b - \mu_{\alpha_1} - \mu_{\alpha_2} - 2\mu_{s_2}) + \mu_{\alpha_1} \\
&= E(\alpha_1 - \mu_{\alpha_1}|\alpha_1 - \mu_{\alpha_1} + 2s_2 - 2\mu_{s_2}) + \mu_{\alpha_1}
\end{aligned}$$

where $(\alpha_1 - \mu_{\alpha_1}) \sim \mathcal{N}(0,\, \sigma^2_{\alpha_1})$ and $(2s_2 - 2\mu_{s_2}) \sim \mathcal{N}(0,\, 4\sigma^2_{s_2})$. Therefore, $E(\alpha_1 - \mu_{\alpha_1}|\alpha_1 - \mu_{\alpha_1} + 2s_2 - 2\mu_{s_2}) = \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1}+4\sigma^2_{s_2}}(\alpha_1 - \mu_{\alpha_1} + 2s_2 - 2\mu_{s_2}) = \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1}+4\sigma^2_{s_2}}(2x - b - \mu_{\alpha_1} - \mu_{\alpha_2} - 2\mu_{s_2})$,[2] and

$$E(\alpha_1|x) = \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1} + 4\sigma^2_{s_2}}(2x - b - \mu_{\alpha_1} - \mu_{\alpha_2} - 2\mu_{s_2}) + \mu_{\alpha_1}.$$

Combining this expression with the first-order condition of the maximization problem for $P_2$ allows us to calculate $y$ in expectation,

$$E(y|x) = \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1} + 4\sigma^2_{s_2}}x + \frac{1}{2}(\mu_{\alpha_2} + \mu_{\alpha_1} + 2\mu_{s_1} - \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1} + 4\sigma^2_{s_2}}(b + \mu_{\alpha_1} + \mu_{\alpha_2} + 2\mu_{s_2}))$$

and thus

$$\begin{aligned}
b &= \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1} + 4\sigma^2_{s_2}}, \\
z &= \frac{1}{2}(\mu_{\alpha_2} + \mu_{\alpha_1} + 2\mu_{s_1} - \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1} + 4\sigma^2_{s_2}}(b + \mu_{\alpha_1} + \mu_{\alpha_2} + 2\mu_{s_2})).
\end{aligned}$$

Given $b$, $z$, and the expression $E(\alpha_1|x)$, the equilibrium reports can be com-

[2] Given independent $X \sim N(\mu_X, \sigma^2_X)$, $Y \sim N(\mu_Y, \sigma^2_Y)$, and $Z = X + Y$. The target is to compute $E[X|Z = z]$. In a bivariate normal distribution,

$$E[X|Z = z] = E[X] + \rho_{XZ} \times \frac{\sigma_X}{\sigma_Z} \times (z - E[Z])$$

where $\sigma_Z = \sqrt{\sigma^2_X + \sigma^2_Y}$ and $\rho_{XZ} = \frac{Cov(X,Z)}{\sigma_X\sigma_Z} = \frac{Cov(X,X+Y)}{\sigma_X\sigma_Z} = \frac{Var(X)}{\sigma_X\sigma_Z} = \frac{\sigma_X}{\sqrt{\sigma^2_X+\sigma^2_Y}}$. Substituting the terms, we derive

$$E[X|Z = z] = \mu_X + \frac{\sigma^2_X}{\sigma^2_X + \sigma^2_Y} \times (z - \mu_X - \mu_Y).$$

puted as follows,

$$\begin{cases} x^* = \frac{1}{2}(\alpha_1 + b + \mu_{\alpha_2} + 2s_2) \\ y^* = bx^* + z + \frac{1}{2}(\alpha_2 - \mu_{\alpha_2}) + (s_1 - \mu_{s_1}) \end{cases} \tag{2}$$

where $b = \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1}+4\sigma^2_{s_2}}$, and $z = \frac{1}{2}(\mu_{\alpha_2} + \mu_{\alpha_1} + 2\mu_{s_1} - \frac{\sigma^2_{\alpha_1}}{\sigma^2_{\alpha_1}+4\sigma^2_{s_2}}(b + \mu_{\alpha_1} + \mu_{\alpha_2} + 2\mu_{s_2}))$.

## Theoretical Predictions

When both players have identical reciprocity coefficient distributions, and draw from the same distributions, and all variances equal to 1, i.e., $\mu_{\alpha_1} = \mu_{\alpha_2} = \mu_\alpha$, $\mu_{s_1} = \mu_{s_2} = \mu_s$ and $\delta_{\alpha_1} = \delta_{\alpha_2} = \delta_{s_1} = \delta_{s_2} = 1$, $b$ equals to $\frac{1}{5}$ and $z$ becomes $\frac{4}{5}(\mu_\alpha + \mu_s) - \frac{1}{50}$. Therefore, the equilibrium reports in (1) and (2) reduce to,

$$\begin{cases} x^*_{si} = \frac{1}{2}(\alpha_1 + \mu_\alpha + 2s_2) \\ y^*_{si} = \frac{1}{2}(\alpha_2 + \mu_\alpha + 2s_1) \end{cases} \qquad \begin{cases} x^*_{se} = \frac{1}{2}(\alpha_1 + \mu_\alpha + 2s_2 + \frac{1}{5}) \\ y^*_{se} = \frac{1}{2}(\alpha_2 + \mu_\alpha + 2s_1) \end{cases} \tag{3}$$

where $(x^*_{si}, y^*_{si})$ is the equilibrium for the simultaneous setting and $(x^*_{se}, y^*_{se})$ is for the sequential setting. That is, $P_2$ believes that $P_1$ inflated her report in the sequential setting by $\frac{1}{5}$ compared to the simultaneous one, and $P_1$ optimally inflates her report by the same amount, leading in both cases to an accurate assessment of the $P_1$'s reciprocity coefficient, and subsequently to the same $y^*$ conditional on $s_1$.

In expectation, we have $E(x^*_{si}) = E(y^*_{si}) = E(y^*_{se}) = \mu_\alpha + \mu_s$, and $E(x^*_{si}) < E(x^*_{se}) = \mu_\alpha + \mu_s + \frac{1}{10}$. Note that all equilibrium reports are higher than $\mu_s$.

**Prediction 1.** *All movers in both settings over-report.*

**Prediction 2.** *The second mover in the sequential setting reports the same in expectation as the players in the simultaneous setting.*

**Prediction 3.** *The first mover in the sequential setting reports more in expectation comparing to the players in the simultaneous setting.*

# 3 Experimental Design

The general setting is based on the dice-rolling paradigm of Fischbacher and Föllmi-Heusi (2013), which utilises events with known distributions (such as dice-rolling), and compares the self-reported distributions with distributions under truth-telling to detect lies at the aggregate level. We instructed the participants to roll a dice (physically or via an online method) and to report the outcome, which shall determine the payments of another participant. If they report numbers 1, 2, 3, 4 or 5, the other participant receives an equivalent payment in British pounds (e.g., 1 representing £1), but if they report a 6, the payment is £0. Our four experimental conditions are as follows, and all of them are one-shot.

***Non-reciprocal (NR)*** In a large group, everyone observes a dice-roll and reports a number that determines the payoff of someone else. Participants do not know who the "someone else" is. However, they do know that this "someone else" is not the person whose reports will count for themselves.

***Simultaneous and Reciprocal (SiR)*** Subjects are randomly paired into groups of two. In a given group with subject $A$ and subject $B$, player $A$ observes a dice-roll and reports a number that determines $B$'s payoff, and in the meantime, player $B$ observes another dice-roll and reports a number that determines $A$'s payoff. Note that both of them report for the other person before observing what the other person reported for them, i.e., players move simultaneously.

***Sequential and Reciprocal (SeR)*** Subjects are randomly paired into groups of two. In a group with subject $A$ and subject $B$, $A$ is the first mover, who first observes a dice-roll and then reports a number that determines $B$'s payoff. Then, the second mover $B$ observes $A$'s report, as well as a dice-roll and then reports a number that determines $A$'s payoff.

***Sequential and Generalised-reciprocal (SeGR)*** Subjects are randomly paired into groups of three, namely subject $A$, $B$ and $C$. $A$ is the first mover, who observes a dice-roll and reports a number that determines $B$'s payoff. Then, the second mover $B$ observes $A$'s report, as well as a dice-roll and then reports a number for $C$.

The first two conditions, NR (Control 1) and SiR (Control 2), serve as two controls corresponding to the "simultaneous setting" of the model. Regarding the peer-review process in large fields, reviewers generally do not know who will be assessing them when they are authors, and the likelihood of being assessed by someone they assessed before is low. NR is the natural point of departure. Further, people may instinctively extrapolate from current interaction into the future, expecting relationships to last. We shall isolate the effect of indirect reciprocity, namely the tendency to treat others well in small groups, comparing it with the purely anonymous environment in NR. It is possible that the feeling of assessing someone who assesses you increases the tendency to be lenient. Therefore, SiR is added as another control.

Our main experimental condition is SeR (Treatment 1), in which there is the prospect of tangible advantages of being lenient because a subject will be assessed by the same person whom she treated leniently. In this sequential setting, reciprocity plays a role. The first mover knows that the second mover will observe her report before reporting for her, so she may try to be nice to the second mover, hoping they may pay her back. The second mover observes the first mover's report and decides whether she reciprocates or retaliates.

Finally, we add condition SeGR (Treatment 2) as an extension, which does not test the current model. It extends the environment to one where the distribution of altruism parameters in the population is unknown, and observing kind behaviour by the first mover may allow the second mover to infer a "better" distribution, and hence treat others better. In small fields, a "culture" of being lenient to others may develop, rather than directly reciprocating lenience.

### Our Preregistered Hypotheses

We first turn our attention to the hypotheses in our preregistration. They will be the main confirmatory hypotheses and other results will be of an exploratory nature.

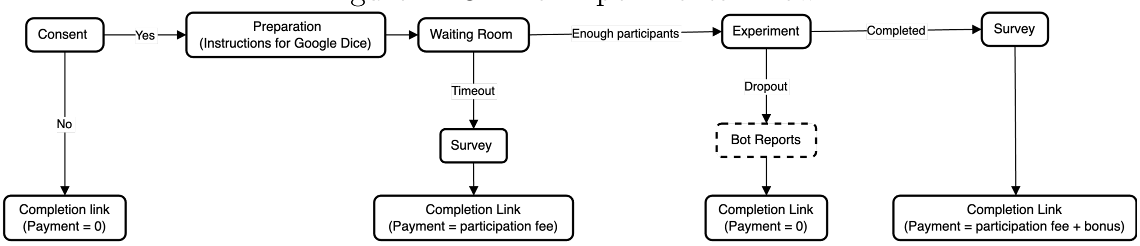
**The Main Hypothesis:** The main hypothesis of our study concerns differences across the reported distributions of participants in Control Condition 1 (NR) and the first movers in Treatment 1(SeR), as per the predictions of our formal model.

**Ancillary Hypotheses:** Our ancillary hypotheses are that belonging to a group matters for reporting behaviour (reporting behaviour across the two control conditions - NR vs. SiR - differs) and that 'generalized reciprocity' matters in that reporting behaviour of 'B players' in Treatment 2 (SeGR) will depend on the reports they received from 'A players'.

## Implementation of the Experiments

We programmed the experiments using oTree (Chen et al., 2016) and conducted these sessions online via the Prolific platform. Given the dynamic nature of online experiments, subjects could potentially withdraw at any stage. Moreover, certain experimental conditions necessitated the formation of groups with multiple participants online simultaneously. Hence, our experimental flow displayed differences from standard lab sessions (See Figure 1).

Figure 1: Online Experimental Flow




To begin the experiment, participants log into their Prolific account and click on our oTree link. The first page they see is the "Consent" page, where they choose to attend the experiment ("Yes") or quit ("No"). Selecting "Yes" takes them to the "Preparation" page, where they receive instructions for setting up Google Dice. Selecting "No" redirects them to the "Completion link" with no payment. When the required number of participants is not achieved (for instance, conditions SiR and SeR require an even number of participants online), some subjects may be placed in a waiting room. If the required number is reached within a certain time frame, the experiment begins. Otherwise, waiting room participants are redirected to the survey and compensated with a fixed participation fee.

During the experiment, timeouts are set for all pages to handle potential dropouts. If a participant exceeds the timeout, they are considered a dropout and do not receive any payment. They are then replaced by a bot that reports a random number. The participant who was grouped with the dropout, without being aware of it, receives a bonus based on the bot's report. This data point is excluded from further analysis. Finally, the remaining participants are asked to complete a survey and are informed about their earnings, which are paid via Prolific the following day. The comprehensive set of instructions of the four experimental conditions can be found in Appendix A.

Experimental sessions for the conditions NR, SiR, and SeR took place in September 2023, and sessions for the SeGR condition were held in October 2023. Participants were recruited randomly from the UK and the US populations. The total participants for NR, SiR, SeR, and SeGR conditions were 174, 187, 331, and 508, respectively. However, taking into account attrition (dropouts and participants who grouped with dropouts), and the fact that receivers in the SeGR condition do not report anything, the actual numbers of observations for each condition are 165 (NR), 163 (SiR), 302 (SeR), and 310 (SeGR).

# 4 Results

Some demographic characteristics of the participants are shown in Table 1. There are no particular differences in basic demographics across conditions, which are relatively similar in age, gender composition and religious affiliation. Figure 2 presents the overall average reporting across treatments (separating first from second movers in the sequential conditions). As can be seen, there is some evidence that first movers in the sequential treatments (even the generalised-reciprocal one) report high.[3] Note that in all treatments except NR participants were matched with the person they assessed in a small group (of two or three persons). Interestingly, the lowest reports can be found for the NR treatment, the most anonymous one.

Table 1: Basic Demographics

| | N | Mean Age | Female (%) | No Religion (%) |
|---|---|---|---|---|
| NR | 165 | 39.69 | 54.5 | 57.0 |
| SiR | 163 | 40.09 | 53.4 | 55.2 |
| SeR | 302 | 39.31 | 57.0 | 63.6 |
| SeR: First | 151 | 40.02 | 58.3 | 60.9 |
| SeR: Second | 151 | 38.60 | 55.6 | 66.2 |
| SeGR | 310 | 39.79 | 56.5 | 62.3 |
| SeGR: First | 155 | 41.99 | 57.4 | 57.4 |
| SeGR: Second | 155 | 37.59 | 55.5 | 67.1 |

*Notes* For the Sequential and Generalised-reciprocal (SeGR) treatment, we actually have a total of 465 participants, with 155 participants for each role. However, one-third of these participants are receivers who did not report anything. Therefore, the number of observations we have for SeGR is 310.

In Figure 3a, we can present in detail the reporting behaviour per individual treatment. Across the board, there seems to exist some tendency for subjects to over-report in the direction of reporting 5. However, the model's predictions focus on the behaviour of first movers in the (main) treatment SeR (Sequential and Reciprocal). As can be seen in Figure 3b, there is a high fraction of first-movers that report 5 in treatment SeR. A similarly high fraction can be observed in the second sequential treatment we examine, namely SeGR (Sequential and Generalised

[3]Please note that in our experiment, reporting a dice roll of 6 results in a payoff of 0. The following analysis focuses on the "reported payoff " (ranging from 0 to 5) rather than the dice roll, as in Fischbacher and Föllmi-Heusi (2013).

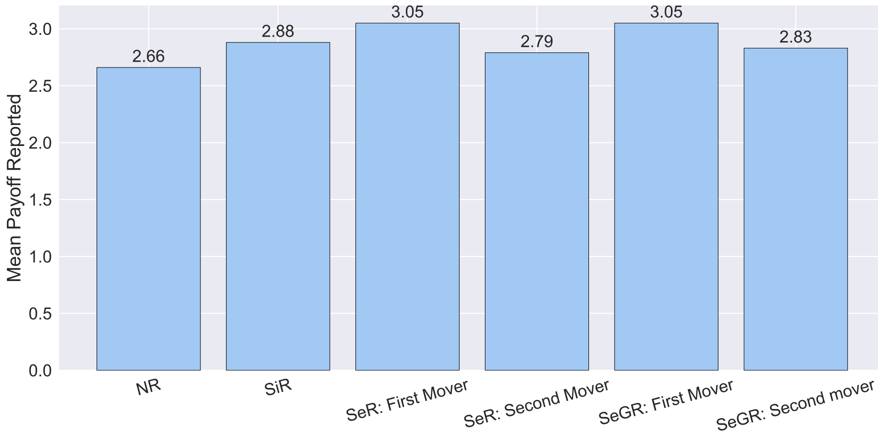



Figure 2: Overall mean reporting for all conditions.

Reciprocal). Juxtaposing the reporting behaviour of first and second movers in the two sequential treatments (with the help of Figure 3c), we observe that while the reporting of first-movers indicates over-reporting of number 5, for second-movers the distribution is more even, with a much weaker tendency to report high. We can now focus in detail on the statistical evidence for our main experimental hypotheses.

Table 2: Whether the Reports are Uniform

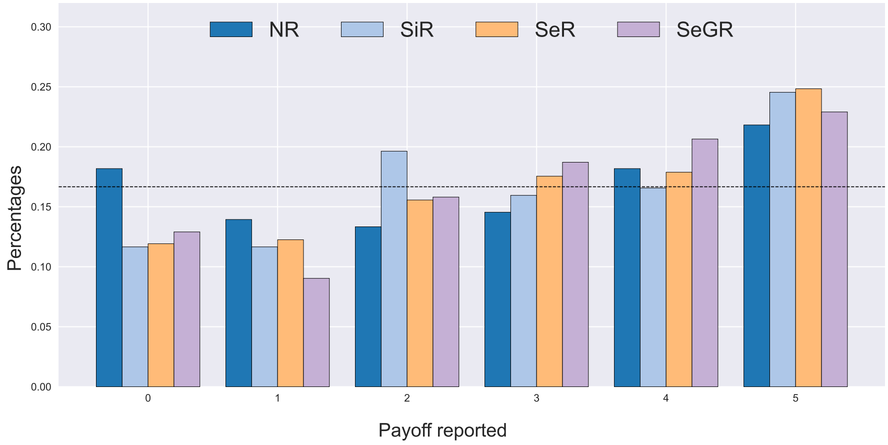
| | **Kolmogorov-Smirnov test** | $\chi^2$ **test** |
|---|---|---|
| **NR** | 0.0000*** | 0.3731 |
| **SiR** | 0.0000*** | 0.0364** |
| **SeR** | 0.0000*** | 0.0011*** |
| **SeR: First** | 0.0000*** | 0.0002*** |
| **SeR: Second** | 0.0000*** | 0.2307 |
| **SeGR** | 0.0000*** | 0.0002*** |
| **SeGR: First** | 0.0000*** | 0.0011*** |
| **SeGR: Second** | 0.0000*** | 0.0216** |

*Notes* Both tests are two-sided. $*p < .1, **p < .05, ***p < .01$

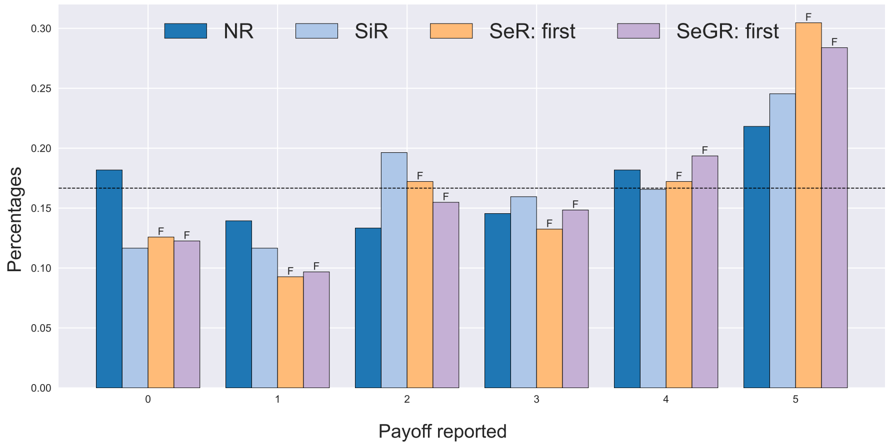
**HYPOTHESIS 1:** *In condition SeR and in condition SiR all movers over-report.*

This hypothesis comes from Prediction 1 of our model. First of all, as can be seen from Table 3 when we pool the first and second movers in SeR, players over-report number five in their reports (p-values of binomial tests are less than 0.05). As Figure 2 shows, in these two conditions, and for all types of movers, average reports exceed the unbiased expected report of 2.5. As Table 2 shows, Kolmogorov - Smirnov tests along with chi-square test reject the hypothesis that for any of our

(a) Pooled




(b) First Movers




(c) Second Movers


Figure 3: Reporting Distributions of All Conditions

*Note:* The dashed black lines represent 1/6. 'F' and 'S' above the bars represents 'First mover' and 'Second mover'.

four treatments the distribution of reports is uniform (p-values are less than 0.05 for any comparison). Accordingly, there is some systematic pattern of divergence from pure honestly driven by the experimental environment, where one can benefit (or harm) others by lying in their report.

Table 3: Share of payoff reported (in percent) and binomial tests

| | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| NR | 18.2 | 13.9 | 13.3 | 14.5 | 18.2 | 21.8*+ |
| SiR | 11.7*-- | 11.7*-- | 19.6 | 16 | 16.6 | 24.5**+++ |
| SeR | 11.9**-- | 12.3**-- | 15.6 | 17.5 | 17.9 | 24.8***+++ |
| SeR: First | 12.6 | 9.3**--- | 17.2 | 13.2 | 17.2 | 30.5***+++ |
| SeR: Second | 11.3*-- | 15.2 | 13.9 | 21.9+ | 18.5 | 19.2 |
| SeGR | 12.9*-- | 9.0***--- | 15.8 | 18.7 | 20.6*++ | 22.9***+++ |
| SeGR: First | 12.3- | 9.7**--- | 15.5 | 14.8 | 19.4 | 28.4***+++ |
| SeGR: Second | 13.5 | 8.4***--- | 16.1 | 22.6*++ | 21.9*+ | 17.4 |

*(a)* Two-sided binomial test, $*p < .1, **p < .05, ***p < .01$
*(b)* One-sided binomial test (greater then 100%/6), $+p < .1, ++p < .05, +++p < .01$
*(c)* One-sided binomial test (less then 100%/6), $-p < .1, --p < .05, ---p < .01$

**HYPOTHESIS 2:** *Participants in SiR and second movers in SeR, report the same.*

Hypothesis 2 directly follows from Prediction 2 of our model. To assess equivalence, we use a Two One-Sided Test (TOST). The null hypothesis is that $|mean(SiR) - mean(SeR : first)| > \delta$, where $\delta$ is the equivalence margin representing the largest acceptable difference between the group means. Using a significance level of 0.05, we conducted a permutation test and determined the smallest $\delta$ to be 0.4. This means that in order to claim $mean(SiR) = mean(SeR : first)$, we must tolerate a difference of at least 0.4. However, this difference is too large given the sample means: SiR reports 2.88 and SeR- first mover reports 2.79.[4] Therefore, we conclude that we do not have significant results for this hypothesis.

**HYPOTHESIS 3:** *In condition SeR, first-movers report on average higher that in condition SiR.*

This hypothesis comes from Prediction 3 of our model, where in the sequential environment first-movers' reports are predicted to be higher than in the simultaneous

[4] At a significance level of 0.1, the smallest $\delta$ is reduced to 0.34, which is still too large. A t-test is also performed, yielding a smallest $\delta$ of 0.331 when $\alpha = 0.1$, and 0.4 when $\alpha = 0.05$. However, we should exercise caution in interpreting the t-test results due to the non-normality of the data.

environment. Table 1 demonstrates that the reported payoffs for SiR and for both movers in SeR are higher on average than the baseline payoff of 2.5. Specifically, SiR has a reported payoff of 2.88, first movers in SeR have a reported payoff of 3.05, and second movers in SeR have a reported payoff of 2.79. The difference between the average report for SiR and for the first mover in SeR is not statistically significant (refer to Table 4).

Table 2 indicates that both the Kolmogorov-Smirnov test and the $\chi^2$ test reject the hypothesis of uniformity for the reports of SiR and the reports of the first mover in SeR.[5] Table 3 displays the reported payoff percentages of participants in all experimental conditions. It also includes binomial tests to check whether the observed percentages differ from 16.7%. The results indicate that participants in SiR and first movers in SeR both significantly over-reported five, the highest payoff.

Table 4: Mean Payoff Reported Differences Between Treatments

| | **Mean Difference** | **K-S** | **Permutation** | **MWU** |
|---|---|---|---|---|
| **(NR vs. SiR)** | -0.2167 | 0.509 | 0.2658 | 0.2905 |
| **(NR vs. SeR: First)** | -0.3858 | 0.3449 | 0.0608* | 0.0537* |
| **(NR vs. SeR: Second)** | -0.1275 | 0.8091 | 0.5094 | 0.5979 |
| **(SiR vs. SeR: First)** | -0.1691 | 0.8556 | 0.3891 | 0.3342 |
| **(SiR vs. SeR: Second)** | 0.0892 | 0.9663 | 0.6681 | 0.6105 |
| **(NR vs. SeGR: First)** | -0.3846 | 0.3468 | 0.0594* | 0.0575* |
| **(NR vs. SeGR: Second)** | -0.1717 | 0.3468 | 0.379 | 0.4944 |
| **(SiR vs. SeGR: First)** | -0.1679 | 0.8422 | 0.3924 | 0.3468 |
| **(SiR vs. SeGR: Second)** | 0.045 | 0.7775 | 0.8153 | 0.7821 |
| **(SeR: First vs. SeGR: First)** | 0.0012 | 1 | 1 | 0.949 |
| **(SeR: Second vs. SeGR: Second)** | -0.0442 | 0.9939 | 0.8318 | 0.8146 |

*(a)* Mean Difference: The mean payoff reported of the first element in brackets minus the mean of the second element.
*(b)* K-S: Kolmogorov–Smirnov Test; Permutation: Fisher-Pitman Permutation Test; MUW: Mann-Whitney U Test
*(c)* All tests are two-sided. $*p < .1, **p < .05, ***p < .01$

**HYPOTHESIS 4:** *First movers in SeR report more than participants in NR.*

Hypothesis 4 corresponds to our first preregistered hypothesis. Getting back to our motivating example of the review process, NR simulates a well-implemented peer review process preserving anonymity, while SeR mimics a process where anonymity cannot be fully preserved in a small field. The mean report for condition NR is 2.66,

[5] For completeness, the reports of the second movers in SeR show significant results according to the Kolmogorov-Smirnov test, but the $\chi^2$ test yields an insignificant result ($p = 0.237$).

while for the first movers of SeR it is equal to 3.05. The results of the Permutation test and MWU test in Table 4 show a significant difference in average payoffs at the 10% significance level, while the K-S test is not significant.

**HYPOTHESIS 5:** *Participants in SiR report more than participants in NR.*

This Hypothesis corresponds to our second preregistered hypothesis. The key difference between SiR and NR lies in the group cohesion participants might feel in SiR, in contrast to the non-reciprocal nature of NR. We hypothesized that SiR participants, due to the sense of belonging to a group, might show more kindness towards one another, potentially yielding higher reports. While we have observed a lower mean report in NR compared to SiR, this difference lacks statistical significance (K-S: $p = 0.509$, Permutation: $p = 0.2658$, MWU: $p = 0.2905$, see Table 4).

**HYPOTHESIS 6:** *There is a positive correlation between the behaviour of 'Player Bs' in and 'Player As' in treatment SeGR.*

This is one of the ancillary hypotheses from our preregistration. As Table 5 shows, the data find no evidence in favour of this hypothesis, as the actual correlation is negative. This indicates that generalised reciprocity does not seem to play a strong role in the behaviour of our participants. It is worth noting that this lack of correlation is consistent with our main model, which assumes only direct reciprocity, but not generalised one.

Table 5: Correlations between the reports of first and second movers

| | beta coef (OLS) | Pearson correlation | Spearman correlation |
|---|---|---|---|
| **SeR** | 1.726* | 0.14* | 0.147* |
| **SeGR** | -0.397 | -0.032 | -0.038 |

*Notes* $*p < .1, **p < .05, ***p < .01$

To summarise our findings: in all conditions the reporting distribution deviates from the uniform one, and average reports exceed the unbiased expectation of 2.5. These effects are more prolonged in the conditions where subjects were matched in a group together with the participants for whom they reported. Differences in average reports were typically not statistically significant in comparisons between

(a) SeR


(b) SeGR

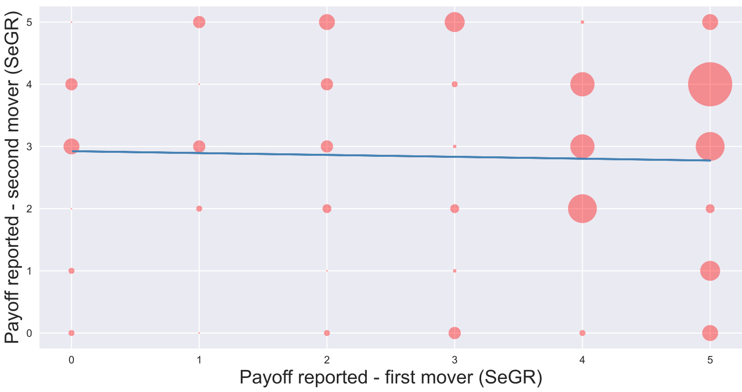



Figure 4: First movers' reports against second movers'

*Note:* The bubble size represents frequency, and large bubbles in the plot are intentionally exaggerated for better visibility. The highest frequency in SeR is for the combination (5, 5), with a frequency of 11. The highest frequency in SeR is for the combination (5, 4), with a frequency of 12. The complete table of all combinations is available in the Appendix B.

treatments.

## 5 Conclusions

We examined an environment that simulates players' assessing each other's performance. We built a model of reciprocity that predicts that, in a sequential setting, the first player will report a higher number than in a simultaneous setting. In addition, the model predicts that in equilibrium, expected reports will be higher than actual draws for every category of movers.

We ran preregistered online experiments with four treatments, drawing form the general UK and US population, to test the predictions of our model, as well as a few other predictions. In our experiments, people appear to over-report large numbers – especially the maximum possible number. This can be interpreted as a tendency to signal that one is acting to benefit the participant they are matched with. Interestingly, this tendency is not only evident in the setting where players assess each other, but also in a generalised sequential setting, where player 1's report may affect the behaviour of player 2 towards a third, unknown person.

With our preregistered large online experiments we find some evidence that the shadow of reciprocity plays a role in an environment where people assess each other. In addition, a model of reciprocity in the spirit of Levine (1998) seems to capture important elements of the environment. Interestingly, the behaviour of the second player in the sequential and reciprocal setting seems to correlate with the report of the first player, indicating that it may be feasible to use signalling of one's type in our setting.

Our overall results imply that people consistently over-report when they have the opportunity to benefit others. However, this tendency is stronger in sequential environments that do not involve anonymity between participants. In addition, people seem to choose extreme levels in the support of reports to signal kindness to others. If our results replicate and generalise (in future studies) this will entail that the type of reciprocity that biases assessments, which we study here, is relevant and especially so in domains that make it hard to preserve anonymity.

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# Appendices

## A Experimental Instructions

### Instructions

---

### ◊ Non-reciprocal

At the beginning of the experiment, you will be paired with a randomly selected other participant from the *Prolific* participant pool. You will determine this participant's payoff by reporting your dice roll.

Meanwhile, there is **another** randomly selected participant, who will determine your payoff by reporting their own dice roll. **Note that *Prolific* has a huge participant pool, and this participant will not be the one you are paired with at the beginning.**

As explained, your report of the dice roll determines some other participant's payment. You can see the exact payoff from the following chart.

| Number rolled | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Resulting payoff | £1 | £2 | £3 | £4 | £5 | £0 |

**You will also receive a £1.50 participation fee.**

Note: **Due to the nature of online experiments, participants may be in different time zones. If they cannot submit their dice roll at the same time as you, you won't know your dice-roll related payment at the end of experiment. We will inform you about your total payment in a couple of days.**

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### ◊ Simultaneous and Reciprocal

At the beginning of the experiment, you will be paired with a randomly selected other participant from the *Prolific* participant pool. You will determine the other participant's payoff by reporting your dice roll. Meanwhile, the other participant will determine your payoff by reporting their own dice roll.

As explained, **your report of your dice roll determines how much the other participant earns. The other participant's report of their dice roll determines how much you earn.**

You can see the exact payoff from the following chart.

| Number rolled | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Resulting payoff | £1 | £2 | £3 | £4 | £5 | £0 |

**You will also receive a £1.50 participation fee.**

---

## ♢ Sequential and Reciprocal

At the beginning of the experiment, you will be paired with a randomly selected other participant from the *Prolific* participant pool. The experiment has two stages, and participants make decisions sequentially:

- In Stage 1, the first mover reports their dice roll. The second mover's payoff is determined by the first mover's report.
- In Stage 2, the second mover sees the first mover's report, reports their own dice roll. This report determines the first mover's payoff.

**Whether you or the other participant moves first is determined randomly.**

As explained, **your report of your dice roll determines how much the other participant earns. The other participant's report of their dice roll determines how much you earn.**

You can see the exact payoff from the following chart.

| Number rolled | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Resulting payoff | £1 | £2 | £3 | £4 | £5 | £0 |

**You will also receive a £1.50 participation fee.**

---

## ♢ Sequential and Generalised-reciprocal

At the beginning of the experiment, you will be part of a randomly selected group of

**3 participants** from the *Prolific* participant pool. The 3 participants are randomly assigned to different roles: ***First Mover***, ***Second Mover*** and ***Receiver***.

The experiment has two stages. The ***First and Second Movers*** make decisions sequentially, and the ***Receiver*** does not need to make any decisions:

- In Stage 1, the ***First Mover*** reports their dice roll. The ***Second Mover***'s payoff is determined by the ***First Mover***'s report.
- In Stage 2, the ***Second Mover*** sees the ***First Mover***'s report, reports their own dice roll. This determines the ***Receiver***'s payoff.

Please note that roles are assigned randomly, and the payments for each role are as follows:

- The ***First mover*** receives a random integer payment between £0 and £5, and they does not know about the exact number until the end of the experiment.
- The report of the ***First mover***'s die roll determines how much the ***Second Mover*** earns.
- The report of the ***Second mover***'s die roll determines how much the ***Receiver*** earns.

You can see the exact payoff from the following chart.

| Number rolled | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Resulting payoff | £1 | £2 | £3 | £4 | £5 | £0 |

**You will also receive a £1.50 participation fee.**

# B Additional Tables

Table B.1: The frequency of reports of first and second movers in SeR and SeGR

| ***first mover*** | ***second mover*** | ***frequency (SeR)*** | ***frequency (SeGR)*** |
|---|---|---|---|
| 0 | 0 | 4 | 3 |
| | 1 | 1 | 3 |
| | 2 | 5 | 1 |
| | 3 | 5 | 6 |
| | 4 | 1 | 5 |
| | 5 | 3 | 1 |
| 1 | 0 | 0 | 1 |
| | 1 | 5 | 0 |
| | 2 | 2 | 3 |
| | 3 | 3 | 5 |
| | 4 | 1 | 1 |
| | 5 | 3 | 5 |
| 2 | 0 | 3 | 3 |
| | 1 | 4 | 1 |
| | 2 | 3 | 4 |
| | 3 | 7 | 5 |
| | 4 | 5 | 5 |
| | 5 | 4 | 6 |
| 3 | 0 | 2 | 5 |
| | 1 | 5 | 2 |
| | 2 | 2 | 4 |
| | 3 | 4 | 2 |
| | 4 | 6 | 3 |
| | 5 | 1 | 7 |
| 4 | 0 | 4 | 3 |
| | 1 | 3 | 0 |
| | 2 | 3 | 9 |
| | 3 | 4 | 8 |
| | 4 | 5 | 8 |
| | 5 | 7 | 2 |
| 5 | 0 | 4 | 6 |
| | 1 | 5 | 7 |
| | 2 | 6 | 4 |
| | 3 | 10 | 9 |
| | 4 | 10 | 12 |
| | 5 | 11 | 6 |